MILITARY SPECIFICATION

CONVERTER, LIQUID TO GASEOUS OXYGEN, AIRCRAFT, TYPE A-3

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of converter for storing and converting 5 liters of liquid oxygen to gaseous oxygen at a pressure of 70 psig for aircrew use, designated type A-3.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-V-9050	Valve, Oxygen, Aircraft, Pressure Relief, General Specification For
MIL-C-9082	Converter, Oxygen, Aircraft, Liquid To Gaseous, General Specification For

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-471	Maintainability Demonstration

"Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC CASO/LODS, Federal Center, Battle Creek MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter."

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MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Parts And Equipment, Procedures For Packaging And Packing Of

Air Force-Navy Aeronautical

AND10053	Boss, Pipe-Thread, Internal - Standard Dimensions For
AND10387	Drill Sizes And Drilled Hole Tolerance Twist

(Copies of specifications, standards, handbooks, drawings and publications required by the manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The requirements of Specification MIL-C-9082 apply as requirements of this specification, with the exceptions and additions specified herein. In the event the requirements of the two specifications conflict, the requirements of this specification shall govern. This specification makes provision for first article testing.

3.2 Components. The converter shall consist of the following major component in addition to those specified in Specification MIL-C-9082.

3.2.1 Relief valves. The relief valves shall conform to Specification MIL-V-9050, types I and II, and shall be of such size and shape to fit in the configuration of Figures 2, 3, and 4.

3.3 Design. The converter shall have a minimum capacity of 5 liters of liquid oxygen measured at 1 atmosphere, and shall be capable of providing gaseous oxygen at a pressure of 70 psig. The schematic diagram of the converter shall conform to Figure 1. The converter shall conform to dimensions and general design shown on Figures 2, 3, 4, and 5.

3.3.1 Load limits. The following load limits, expressed as "G" loadings on a full converter, shall be used as a basis of design:

Vertical (down)	+14 G's
Vertical (up)	- 6 G's
Lateral	6 G's

3.3.2 Protective cable. If the converter is subject to shattering when hit by gunfire, it shall be protected by 3/32-inch diameter cable.

3.3.3 Wrenching provisions. Wrench flats shall be provided on female pipe thread bosses located on outer shell of the container.

3.4 Performance.

3.4.1 Evaporation loss. The evaporation loss shall not exceed 2.5 pounds of oxygen within any 24-hour period.

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3.4.2 Permanent volumetric expansion. The inner vessel shall be subjected to a hydrostatic pressure of 120 psig.

3.4.3 Delivery rate. The converter shall be capable of delivering 10 liters of gaseous oxygen per minute at a temperature of not colder than 27°F below ambient temperature. The duration of the test shall be at least 2 hours. The operating pressure shall remain between 65 and 75 psig. The unit shall be capable of delivering 20 liters per minute of gaseous oxygen with no temperature limits for approximately 1 hour without the operating pressure decreasing below 65 psig.

3.4.4 Filling time. Filling time shall not exceed 5 minutes.

3.4.5 Buildup time. Buildup time shall not exceed 10 minutes.

3.4.6 Final assembly pressure. The complete assembly shall be able to withstand 85-psig pressure without leakage.

3.4.7 Low-temperature exposure. The converter shall be exposed to a temperature of not warmer than -65°F for a period of 3 hours. Immediately after this exposure, and while at this temperature, the converter shall function normally, and shall deliver the rated amount of gaseous oxygen for a period of 1 hour. An intermittent increase or decrease of pressure by an additional 5 psig is allowable in the low-temperature range, provided these variances do not exceed a 2-minute interval for any one pressure cycle.

3.4.8 High-temperature exposure. The converter shall be exposed to a temperature of at least 160°F for a period of 3 hours. Immediately after this exposure, and while at this temperature, the converter shall function normally, and shall deliver the rated amount of gaseous oxygen for 1 hour. An intermittent increase or decrease of pressure by an additional 5 psig is allowable in the high-temperature range, provided these variances do not exceed a 2-minute interval of any one pressure cycle.

3.5 Weight. The weight of the complete converter shall not exceed 18 pounds.

* 3.6 Maintainability. In addition to the requirements in MIL-C-9082, the Mean Corrective Maintenance Time (Mct, as defined in MIL-STD-471) of the unit shall not be greater than 30 minutes when repaired by a maintenance technician of skill level 5 or equivalent. The consumers risk for the maintainability test shall be ten percent.

* 3.7 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. The inspection and first article test shall be in accordance with the inspection and qualification tests described in Specification MIL-C-9082.

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* 4.2 Reliability test. One first article sample of the equipment shall be tested by the contractor for reliability in accordance with Test Plan I of MIL-STD-781. A specified MTBF of 300 hours shall be used for the test. Recording data handling and reporting shall be in accordance with MIL-STD-781.

* 4.3 Acceptance reliability test. Acceptance reliability test shall be in accordance with Test Plan I of MIL-STD-781. The specified MTBF of 300 hours shall be used for the test. Recording, data handling and reporting shall be in accordance with MIL-STD-781.

*5. PACKAGING

* 5.1 Preservation-Packaging. Preservation-packaging shall be in accordance with MIL-STD-794 level A or C as specified by the procuring activity.

* 5.1.1 Level A. Each converter pressurized to a maximum of 25 lbs PSI shall be preserved in accordance with Method III of MIL-P-116. The unit container shall conform to PPP-B-636, type C/F, style RSC, class V3c. The closure shall be accomplished in accordance with the procedure specified in the appendix of the applicable specification.

* 5.1.2 Level C. Each unit shall be packaged in a manner that will prevent physical damage during transit from the supply source to the first receiving activity for immediate use.

* 5.2 Packing. Unless otherwise specified by the procuring activity, the items shall be provided with packing which will afford adequate protection against damage during shipment, handling, intermediate storage and world-wide redistribution.

* 5.2.1 Level A. Items packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, styles A and B class overseas. As far as practical, exterior containers shall be uniform in shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities, unless otherwise specified by the procuring activity.

* 5.2.2 Level B. Items packaged as specified in 5.1.1 shall be packed in fiberboard containers conforming to style RSC, class weather resistant, grade V35 or V3C, of PPP-B-636. Containers shall be of uniform shape size and contain identical quantities. The container closure shall be accomplished in accordance with Method II, 30.3.1.2 of the appendix to PPP-B-636.

* 5.2.3 Level C. The packaged items shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

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* 5.3 Marking. All unit and shipping containers shall be marked in accordance with MIL-STD-129. In addition, the following precautionary marking shall appear on the unit and shipping containers:

OXYGEN COMPONENTS-ALL OIL, GREASE,
SHOP RESIDUE, OR OTHER CONTAMINANTS
HAVE BEEN REMOVED.

DO NOT OPEN UNTIL READY FOR USE.

6. NOTES

6.1 General. The instructions contained in Section 6 of Specification MIL-C-9082 are applicable to and form a part of this specification. In the event the requirements of the two specifications conflict, the instructions in this specification shall govern.

6.2 Intended use. The converter covered by this specification is intended to store 5 liters of liquid oxygen and convert it to gaseous oxygen at a pressure of 70 psig, as required by the aircrew during flight.

6.3 First article. When a first article is required, the item will be inspected and tested in accordance with Section 4 of this document. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article.

6.4 Changes from the previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 99

Project number:
1660-F451

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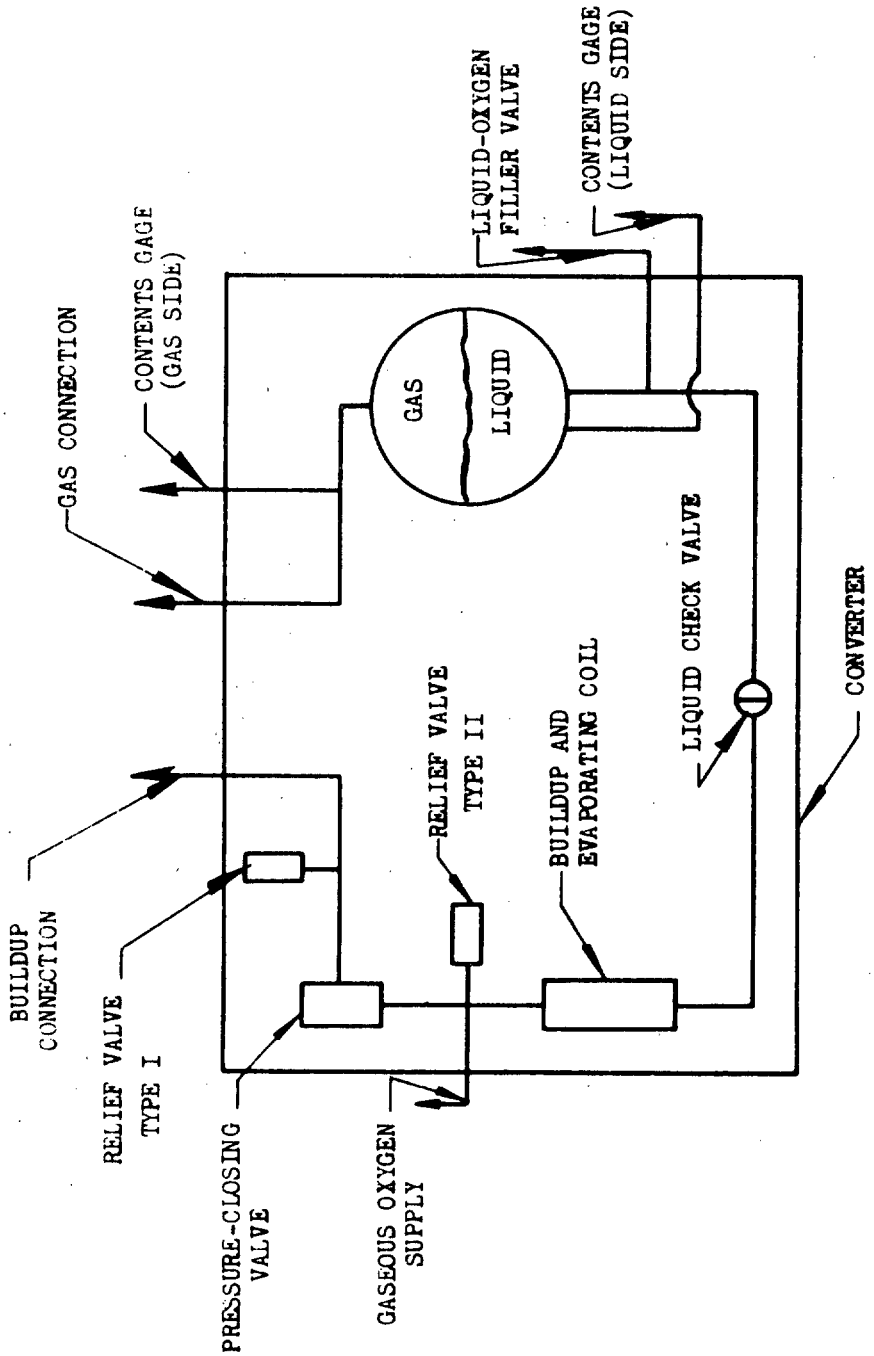
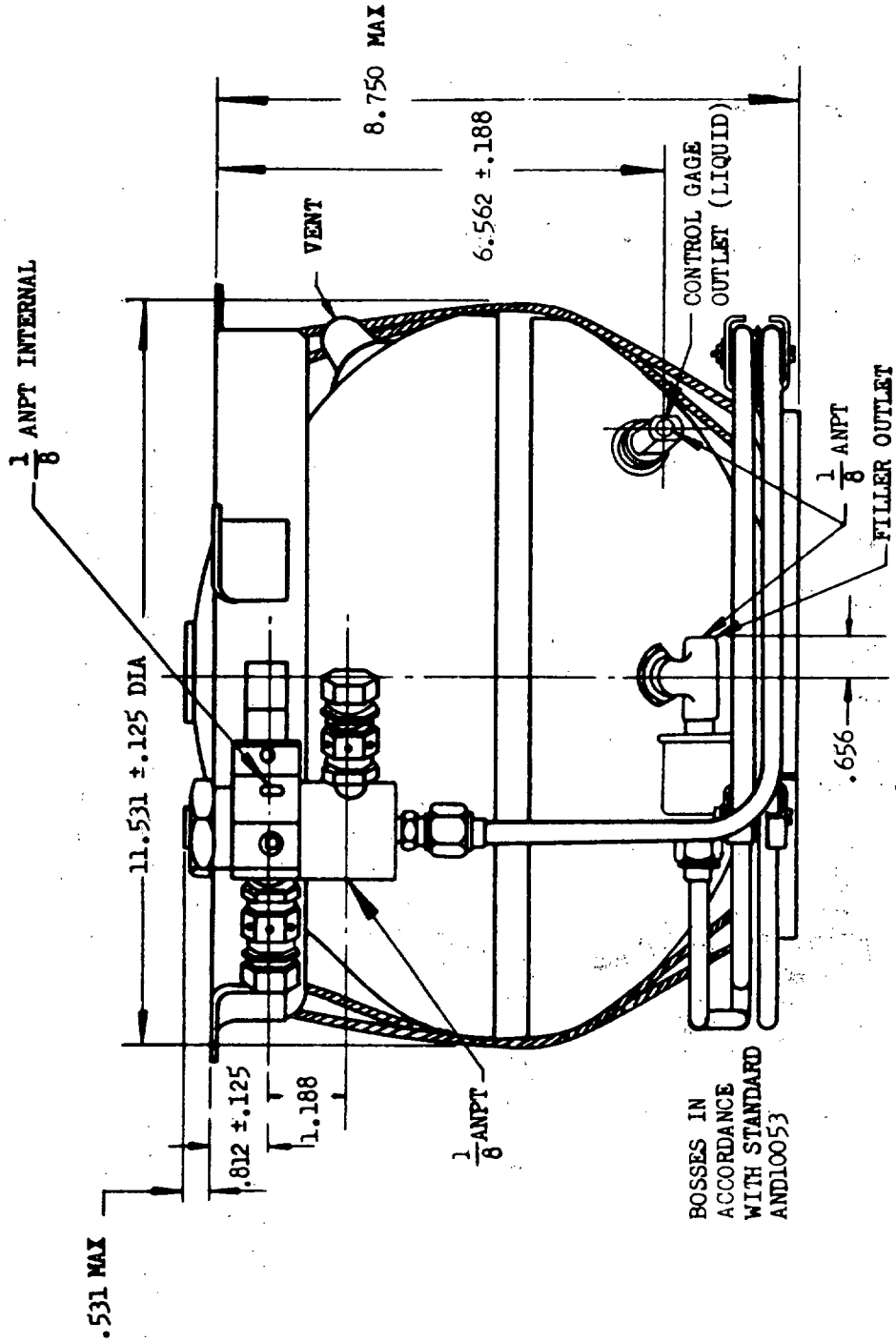


FIGURE 1. Schematic diagram of 5-liter converter.

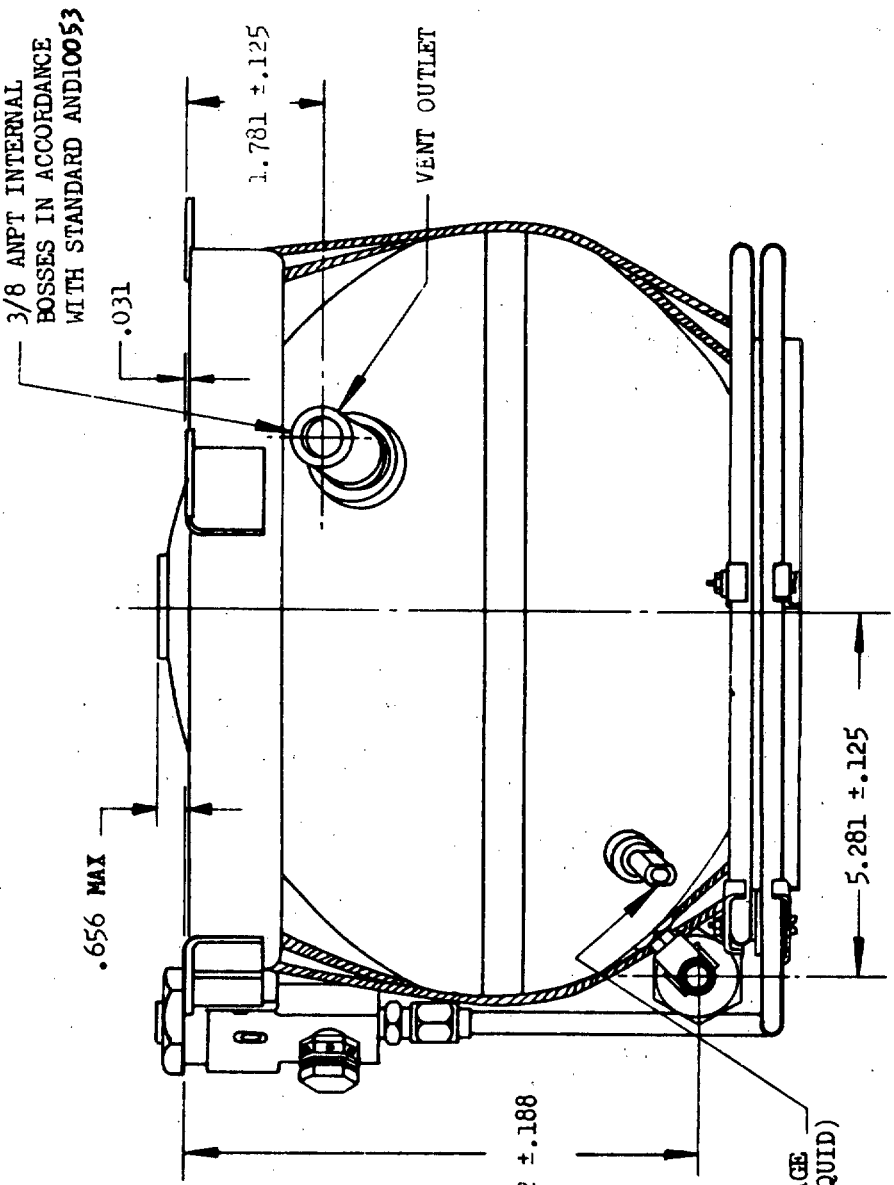
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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.016.

FIGURE 2. General design, front view.

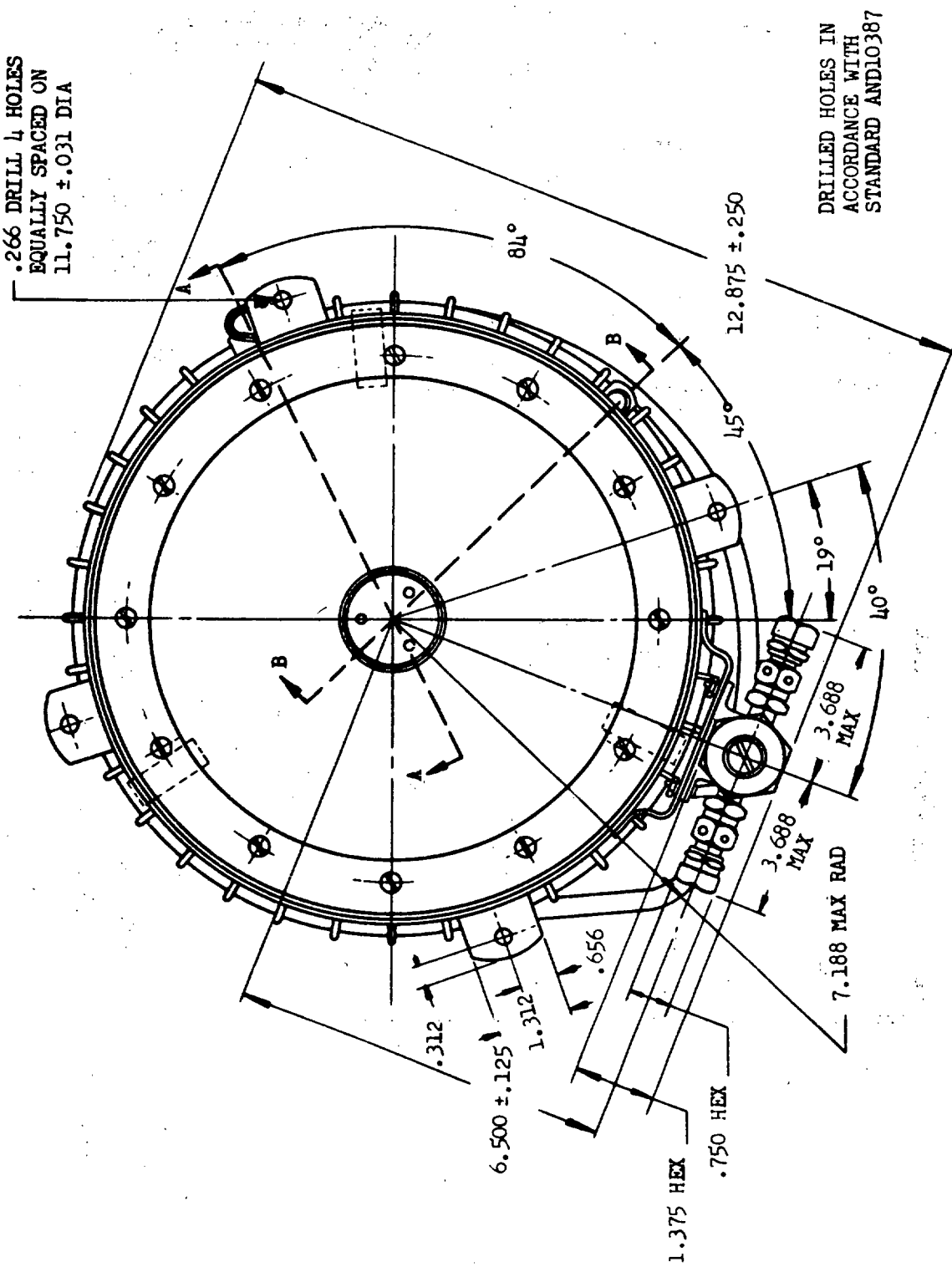
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FIGURE 3. General design, right side view.

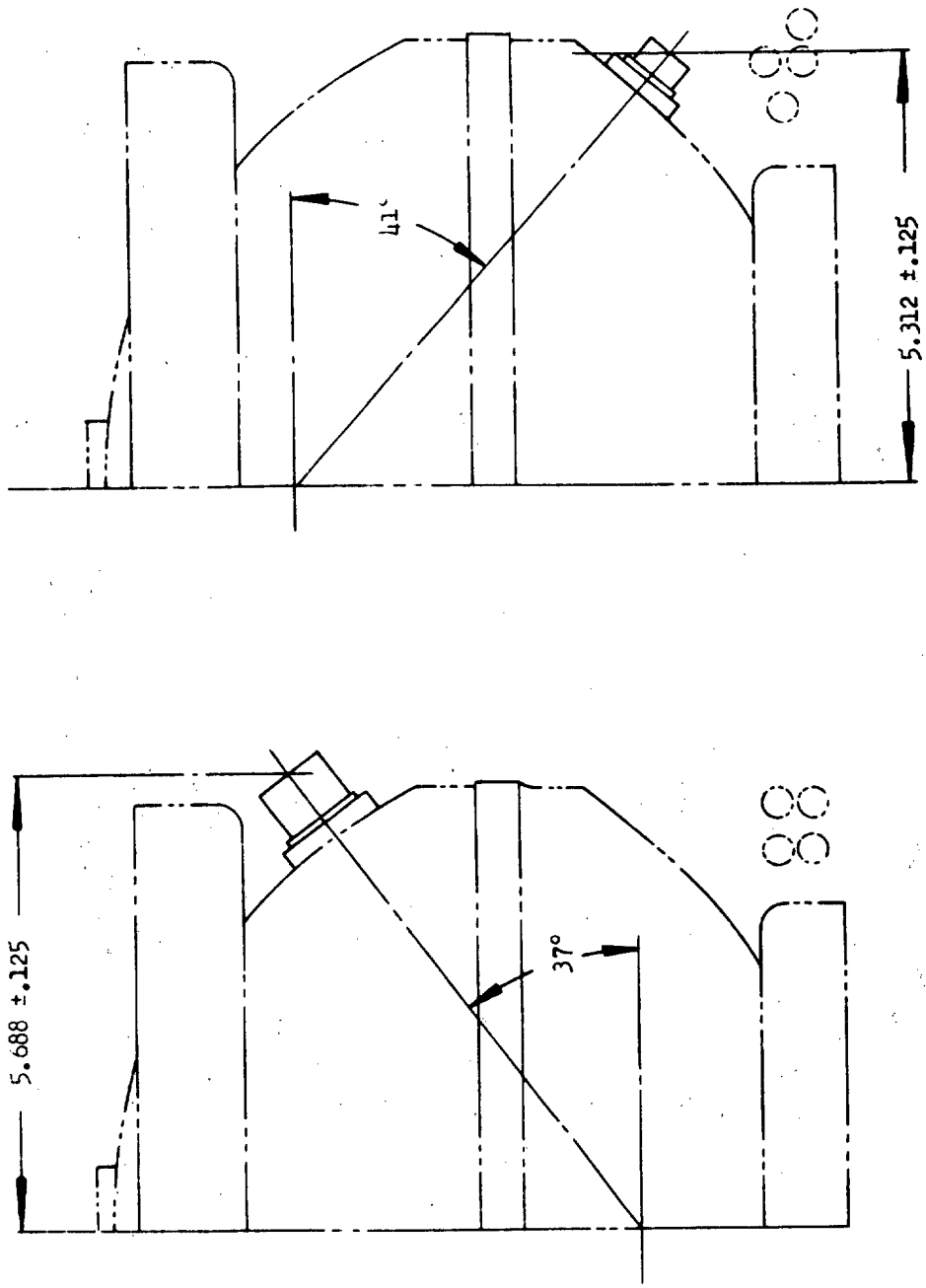
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FIGURE 4. General design, top view.

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FIGURE 5. General design.

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